Exercise 1: SQL fundamental

1 Write a SQL query to retrieve all columns from the employees table

```
Select *
from employees;
```

Id	first-name	last-name	department	Salary	hire-date	City
1	John	Doe	IT	55000	2018-06-15	NewYor
2	Jane	Smith	HR	48000	2019-07-20	Chicag
3	Mike	Johnson	finance	6000	2017-09-30	LosAn
4	Sarah	Brown	IT	53000	2021-03-25	NewYo
5	David	White	marketing	52000	2016-04-10	SanFra
6	Emily	Davis	IT	62000	2015-02-14	Chica
7	Robert	Wilson	finance	59000	2019-10-01	Hous
8	Jessica	Moore	HR	51000	2018-07-22	Los
9	Daniel	Clark	marketing	53000	2022-06-01	Chica
10	Laura	Hall	IT	5000	2020-08-10	San

~~Write a SQL query to find all employees' first and last names, ordered by salary in descending order.~~

2 Write a SQL query to retrieve all the unique departm

```
Select DISTINCT department
From employees
```

Department
IT
HR
finance
marketing

3 Write a SQL query to retrieve all employees' first and last name, ordered by salary in descending order.

```
SELECT first_name,
       last_name
FROM employees
ORDER BY salary DESC;
```

First_name	last_name
Emily	Davis
Mike	Johnson
Robert	Wilson
John	Doe
Daniel	Clark
Sarah	Brown
David	White
Jessica	Moore
Laura	Hall
Jane	Smith

~~Write a SQL query to retrieve top 5 highest-paid employees~~

id	first_name	last_name	department	salary	hire_date	city
6	Emily	Davis	IT	62000	2015-02-14	Chicago
3	Mike	Johnson	finance	6000	2017-09-30	LA
7	Robert	Wilson	finance	59000	2019-10-01	Houston
1	John	Doe	IT	55000	2018-06-10	NY
9	Daniel	Clark	marketing	53000	2022-10-1	Chicago

Write a SQL query to find employees who work in the IT department

Write a SQL query to retrieve the top 5 highest-paid employees

```
SELECT *
FROM employees
ORDER BY Salary DESC
LIMIT 5;
```

id	first_name	last_name	department	salary	hire_date	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
1	John	Doe	IT	55000	2018-06-15	New York
9	Daniel	Clark	Marketing	53000	2022-08-01	Chicago

Write a SQL query to find employees who work in the IT department

```
SELECT *
FROM employees
WHERE department = IT;
```

id	first_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
10	Laura	Hall	IT	50000	2020-12-10	San Francisco

Write an SQL query to find employees who work in the finance department AND have salary greater than 58000

SELECT *
FROM employees
WHERE department = 'finance' AND Salary > $8000;

Id	first_name	last_name	department	Salary	Hiredate	City
3	mike	Johnson	finance	60000	2017-09-30	Los Angeles

Write a SQL query to find employees who work in the HR department or the marketing department

SELECT *
FROM employees
WHERE department = 'HR' or department = marketing;

Id	first_name	last_name	department	Salary	Hiredate	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
5	David	White	marketing	52000	2016-04-10	San francisco
8	Jessica	moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	marketing	53000	2022-06-01	Chicago

Write a SQL query to find employees who do not work in the IT department

SELECT *
from employees
WHERE department != IT;

Id	first_name	last_name	department	Salary	Hire_date	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	finance	60000	2017-09-30	Los Angeles
5	David	White	marketing	5200	2016-04-10	San Francisco
7	Robert	Wilson	finance	59000	2021-10-01	Houston
8	Jessica	moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	marketing	53000	2022-06-01	Chicago

Write an SQL query to find employees who are in the HR, IT or Finance departments.

```
SELECT*
from employees
WHERE department IN ('HR', 'IT', 'finance')
```

id	first-name	last-name	department	Salary	hire date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	mike	Johnson	finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	52000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	finance	59000	2019-10-01	Houston
8	Jessica	moore	HR	51000	2018-05-22	Los Angeles
10	Laura	Hall	IT	50000	2020-08-10	San-Fran

Write a SQL query to find employees who are in the IT department, have a salary greater than 50 000 and are located in New York

```
SELECT*
from employees
WHERE depart='IT' AND Salary > 50000 AND City='New York'
```

id	first-name	last-name	department	Salary	hire date	City
1	John	Doe	IT	55000	2018-06-15	New York

Write a SQL query to retrieve the first and last names of employees who work in the finance or marketing departments earn more than 52000 and order the results by Salary in descending order

```
SELECT first_name,
       last_name
FROM employees
WHERE (department = 'finance' OR department = 'marketing')
AND Salary > 5200
ORDER BY Salary DESC;
```

first_name	last_name
Mike	Johnson
Robert	Wilson
Daniel	Clark

Write a SQL query to find all the unique cities where employees work, excluding those in the IT & HR department

```
SELECT DISTINCT City
FROM employees
WHERE department NOT IN ('IT', 'HR');
```

City
Los Angeles
San francisco
Houston

Write a SQL query to retrieve employees who are not in the finance department, have a salary greater than 50 0000 and order the results by hire date in ascending order

```
SELECT *
FROM employees
WHERE department != 'finance' AND Salary > 50000
ORDER BY hire_date ASC;
```

id	first_name	last_name	department	salary	hire_date	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
9	Daniel	Clark	marketing	53000	2022-06-01	Chicago

Write a SQL query to find the first 3 employees who work either Chicago or Los Angeles and belong to the IT or marketing department

```
SELECT *
FROM employees
WHERE city IN ('Chicago', 'Los Angeles') AND depart IN ('IT', 'marketing')
Limit 3;
```

id	first_name	last_name	departament	salary	hire_date	city
6	Emily	Davis	IT	62000	2015-02-14	Chicago
9	Daniel	Clark	marketing	53000	2022-06-01	Chicago